



Ability to Decipher Emotional Expressions in Typically Developing Bangladeshi 3-7 Years-Olds

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ARTICLE INFO

Article History:

Received: 10th June 2026

Accepted: 7th August 2026

Keywords:

*Emotional Recognition,
Bangladeshi Children, Early
Childhood Development,
Socio-cultural Influence,
Emotional Intelligence.*

JEL Classification:

I10, I12, I18, I30, I31

ABSTRACT

This study examines the emotional recognition abilities of typically developing Bangladeshi children aged 3 to 7, addressing a gap in understanding emotional development within Bangladesh's collectivist cultural framework. The primary objective was to assess children's capacity to decode emotional expressions across various tasks. A purposive sample of 100 children (58 females, 42 males) from four mainstream schools participated in this experimental, cross-sectional study. Data were collected using a culturally tailored tool comprising eight emotional recognition tasks: Personal Narrative, Picture Identification, Pretend Play, Emoji Identification, Coloring Emoji Identification, Sally-Anne False Belief, Emotional Game, and Emotional Sorting. Performance was measured using a scoring system with a maximum total of 40 points. Results indicated that children demonstrated advanced emotional recognition skills, with an average score of 38.65/40, reflecting robust emotional understanding. These findings highlight the role of cultural influences in emotional recognition and underscore the need for culturally adapted emotional literacy programs to enhance socio-emotional development in Bangladeshi children.

1. Introduction

The ability to recognize and interpret emotional expressions represents a fundamental aspect of human communication, enabling individuals to convey and decode affective states, intentions, and social cues through non-verbal signals, including facial expressions, gestures, and vocal tone. This skill is particularly critical during early childhood, as it underpins social interactions, relationship-building, and self-regulation, thereby forming the foundation for emotional and social competence in later developmental stages. Proficient emotional recognition contributes significantly to children's capacity to engage effectively in social contexts, regulate their emotions, and establish healthy interpersonal relationships, which, in turn, supports their psychological well-being and academic achievement. Extant literature indicates that even infants possess the rudimentary ability to differentiate basic emotions such as happiness, sadness, and anger. As children grow, their emotional interpretation skills become increasingly accurate and nuanced. Typically developing children exhibit considerable improvements in recognizing a broad spectrum of emotional states, ranging from basic to complex, highlighting the importance of fostering emotional intelligence from an early age (Bedford et al., 2021). Moreover, early competence in emotion recognition has been consistently associated with enhanced social adaptability and behavioral regulation, emphasizing its role in overall emotional well-being and developmental trajectories (Negrão et al., 2021).

2. Literature Review and Theoretical Framework Development

The cultural context of Bangladesh, characterized by strong familial bonds, collectivist values, and societal norms that prioritize communal harmony over individual expression, offers a unique environment for examining emotional recognition development. Cultural factors are known to significantly influence emotional interpretation and expression, shaping how children perceive and respond to various affective cues. Despite this recognition, there remains a conspicuous lack of research dedicated to understanding emotional recognition among Bangladeshi children.

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This research gap is particularly evident in studies focused on typically developing children aged 3–7 years, whose emotional recognition abilities are likely influenced by familial practices, cultural expectations, and socioeconomic conditions. Emotional intelligence, encompassing abilities such as empathy, self-regulation, and social awareness, is intrinsically linked to emotional recognition. The accurate interpretation of emotional signals often serves as a precursor to successful social interactions, conflict resolution, and emotional resilience. Therefore, investigating how Bangladeshi children develop these skills is essential for informing educational strategies and psychological interventions tailored to this demographic.

Existing studies suggest that children's emotional recognition skills improve with age and are influenced by factors such as parental guidance, peer interaction, and educational experiences (Herba & Phillips, 2004). Neurodevelopmental conditions, such as autism spectrum disorders (ASD), also affect emotional recognition abilities, highlighting the importance of tailored interventions for affected populations (Golan, Baron-Cohen, & Hill, 2006). However, most research in this domain has focused on Western populations, with minimal exploration of how cultural and socioeconomic factors in Bangladesh shape emotional recognition development. Addressing this research gap is crucial for enhancing our understanding of how typically developing Bangladeshi children decode emotional expressions and how these abilities are influenced by their unique cultural context.

Theoretical frameworks underpinning this study include cognitive developmental theories proposed by Piaget and Vygotsky, which emphasize the interplay between age, gender, and sociocultural factors in the development of emotional recognition skills. Additionally, foundational theories such as Darwin's Developmental Perspective (Darwin, 1965), Ekman's Theory of Basic Emotions (Ekman, 1992), Bandura's Social Learning Theory (Bandura & Walters, 1977), and Bowlby's Attachment Theory (Bowlby, Ainsworth, & Bretherton, 1992) provide essential insights into how emotional recognition emerges and matures over time.

Despite valuable contributions from prior studies, the literature remains predominantly focused on Western contexts, lacking culturally nuanced assessments relevant to Bangladeshi children. The absence of appropriate assessment tools and culturally sensitive methodologies continues to impede accurate measurement of emotional recognition in this population. Moreover, the potential of longitudinal studies and technological integration in emotional assessment remains largely unexplored.

This study, therefore, aims to bridge these gaps by providing a comprehensive investigation of emotional recognition development among typically developing Bangladeshi children. The findings are expected to inform the design of culturally relevant educational interventions, parental guidance programs, and tailored training for teachers to enhance emotional literacy from an early age. Ultimately, this research endeavors to contribute to a deeper understanding of the developmental pathways of emotional recognition within the distinct sociocultural framework of Bangladesh.

3. Research Objectives

The primary objective of this study is to investigate how Bangladeshi children aged 3–7 years interpret emotional expressions, with particular attention to false belief comprehension, emotion recognition, and intent understanding. By examining these aspects, the study aims to elucidate the roles of familial practices, cultural expectations, and socioeconomic status in shaping emotional recognition abilities. Furthermore, this research seeks to provide valuable insights to educators, psychologists, and policymakers for supporting the social and emotional development of Bangladeshi children.

4. Research Methodology

This study employed an experimental, cross-sectional design underpinned by quantitative research methodologies to assess the ability of Bangladeshi children aged 3 to 7 years to decipher emotional expressions. Data were collected from a purposive sample comprising 100 typically developing children (58 females and 42 males) recruited from four mainstream educational institutions: 24 No Puran Baushia Government Primary School, Mizanur Rahman Chowdhury Bidyaniketon, Bijoy International School, and Prottoy Inclusive English Medium School. The participants were selected to ensure representation of typically developing children within regular school settings, providing a suitable baseline for examining developmental patterns in emotional recognition skills within this demographic. To gather data, an experimental tool specifically designed for the Bangladeshi context, titled "Ability to Decipher Emotional Expressions in Typically Developing Bangladeshi 3-7 Years Old," was employed.

This tool comprised eight subsections, each aimed at assessing different dimensions of emotional recognition: Personal Narrative, Picture Identification, Pretend Play, Emoji Identification Task, Coloring Emoji Identification Task, Sally-Anne False Belief Task, Emotional Game, and Emotional Sorting. The scoring system assigned a maximum of five points to each subsection, culminating in a total possible score of 40 points. Performance levels were categorized according to a predefined scale: 32-40 points signified normal proficiency, 24-31 points indicated mild difficulty, 16-23 points reflected moderate difficulty, 8-15 points corresponded to severe difficulty, and 0-7 points denoted profound difficulty. This structured scoring approach enabled a nuanced evaluation of participants' abilities to interpret emotional cues across various scenarios.

The collected data were systematically organized and subjected to quantitative analysis using descriptive statistical tools, including tabular presentations, pie charts, and bar charts, to facilitate the identification of trends and developmental patterns. Ethical considerations were meticulously observed throughout the research process, ensuring the confidentiality and anonymity of participants. Consent was duly obtained for the use of photographs and video recordings, in accordance with ethical guidelines governing research involving children. Nonetheless, this investigation offers valuable insights into the developmental trajectory of emotional recognition abilities among typically developing Bangladeshi children aged 3 to 7, contributing to the broader understanding of socio-emotional development within this cultural context.

Study Design	Data Collection Components (Subsections)	Scoring System	Data Analysis	Ethical Considerations
• Design: Cross-sectional study • Participants: 100 typically developing children (58 females, 42 males) • Age Range: 3 to 7 years • Schools Involved: • 24 No Puran Baushia Government Primary School • Mizanur Rahman Chowdhury Bidyaniketon • Bijoy International School • Prottoy Inclusive English Medium School	• Personal Narrative • Picture Identification • Pretend Play • Emoji Identification Task • Coloring Emoji Identification Task • Sally-Anne False Belief Task • Emotional Game • Emotional Sorting	• Maximum Score: 40 (5 points per subsection) • Performance Categories: • Normal Proficiency: 32–40 points • Mild Difficulty: 24–31 points • Moderate Difficulty: 16–23 points • Severe Difficulty: 8–15 points • Profound Difficulty: 0–7 points	• Quantitative tools used: • Tabular Representation • Pie Charts • Bar Charts	• Consent obtained for photos and video recordings. • Anonymity and confidentiality of participants were strictly maintained.

Table-1: Data Collection Tools and Research Design

5. Results and Discussions

5.1. Demographic information

The dataset for this study includes information gathered from a cohort of 100 typically developing Bangladeshi children, aged 3 to 7, comprising both boys and girls from various socioeconomic backgrounds. Data collection was conducted using eight distinct tasks and questionnaires, each specifically designed to evaluate the children's emotional recognition abilities. Each task contributed up to five points to the cumulative total score of 40, providing a detailed measure of each child's capacity for recognizing emotional expressions. Key variables within the dataset include emotional recognition scores derived from each task and demographic factors such as age, gender, and socioeconomic status. Demographic information, includes the total sample distribution, with 42 boys and 58 girls, and an age range of 3 to 7 years, with a mean age of 5. The age breakdown reveals a fairly even distribution across groups, with 12 children aged 3, 10 aged 4, 16 aged 5, 17 aged 6, and 45 aged 7. Further data analysis and results are provided for each of the eight task-based subsections to give comprehensive insight into children's emotional recognition abilities across the age spectrum.

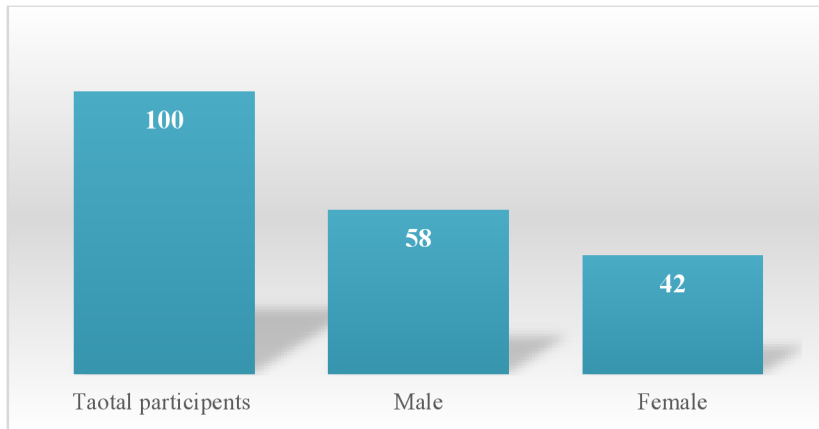


Figure-1: Participant Demographics: Male-to-Female Ratio

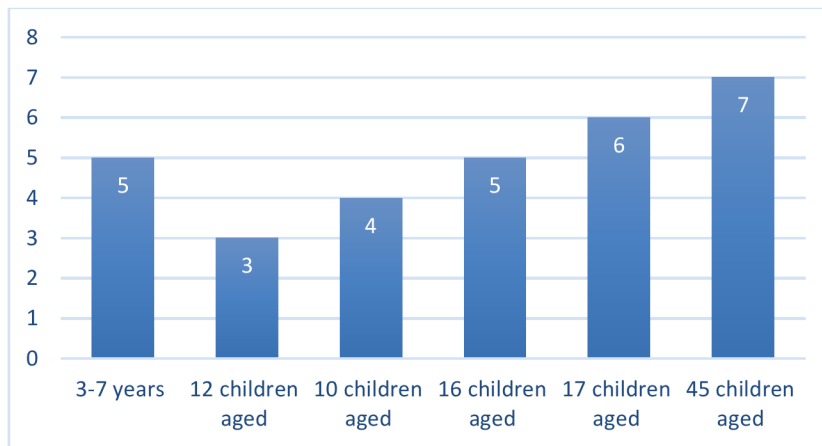


Figure-2: Participant Demographics: Age Range Variation

5.2. Personal Narrative

The assessment comprised five narrative-based items designed to evaluate emotional understanding and interpretive skills. Each of the 100 participating children responded to all five items, with notable results. A substantial majority-92 children-achieved a perfect score of 5 out of 5, accurately responding to all questions. This outcome suggests a high level of emotional recognition ability among most participants. Meanwhile, a smaller subset of 8 children correctly answered 4 questions but were unable to answer one item correctly, resulting in a score of 4 out of 5. Consequently, the overall average performance score across all participants was 4.92 out of 5. This high average reflects the children's strong competency in narrative-based emotional recognition tasks, with minimal variation in interpretive responses among the group.

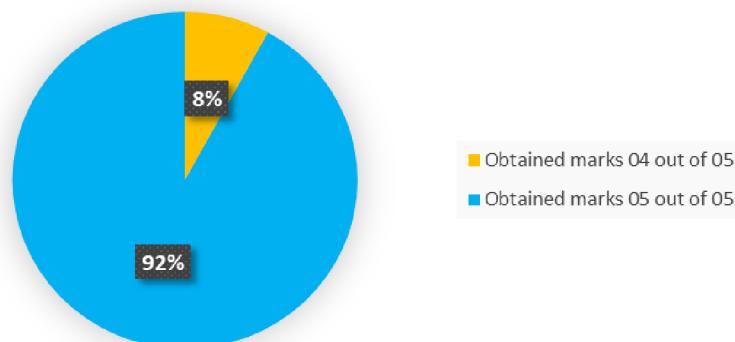


Figure-3: Personal Narrative Task Scores

5.3. Picture Identification Task

The task required presenting six images, each portraying a distinct facial expression representing a basic emotion, to assess the children's ability to recognize emotional cues. A total of 100 children participated, each asked to identify the emotion depicted in each image. Impressively, 85 children successfully identified all six expressions, achieving a perfect score of 5 out of 5. In contrast, 15 children encountered some difficulty, accurately identifying 4 or 5 out of the six expressions, which resulted in an average score of 4 out of 5 for this subgroup. Consequently, the overall performance score across all participants averaged at 4.85 out of 5. This high score demonstrates a strong general ability among the children to recognize and interpret basic emotions, though a small variation in accuracy was observed within the group.

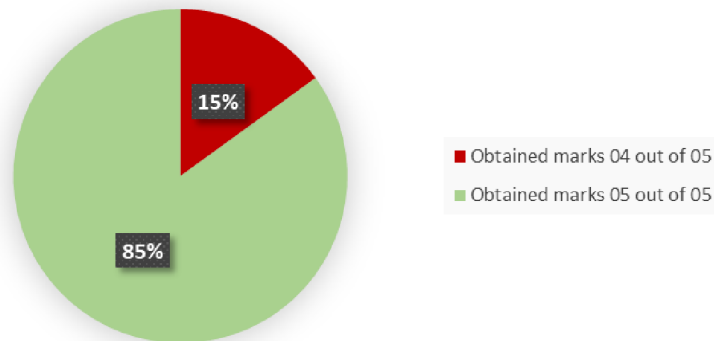


Figure-4: Picture Identification Task Scores

5.4. Pretend Play

In this task, the researcher engaged children in a series of playful activities, first demonstrating various actions and inviting them to join in role-playing scenarios. Additionally, the researcher presented videos showcasing different activities, which the children were then asked to imitate. Every participant accurately enacted each requested activity, demonstrating strong imitation and role-play abilities. Consequently, the task yielded a perfect average score of 5 out of 5, or 100%, underscoring the children's high level of engagement, accuracy, and enthusiasm in replicating all demonstrated scenarios.

5.5. Emoji Identification Task

The assessment involved presenting six distinct emojis, each designed to convey a specific emotional expression. Researchers asked 100 children to identify the emotion represented by each emoji, carefully recording their responses for analysis. Out of the participants, 78 children accurately recognized all six emojis, achieving a perfect score of 5 out of 5. Meanwhile, 22 children correctly identified four out of the six emojis, demonstrating a slight challenge in distinguishing between certain emotions. This resulted in an overall average score of 4.78 out of 5, indicating that while most children exhibited strong emotional recognition skills, there was a minor variance in performance across the group. This high average underscores the children's general aptitude in decoding emotional cues through visual symbols, with only a small gap in complete accuracy.

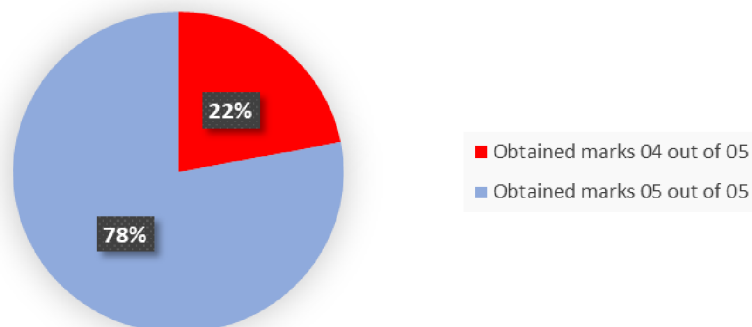


Figure-5: Emoji Identification Task Scores

3.5 Coloring Emoji Identification Task

The task presented six monochrome emojis, each uniquely representing a different emotional expression, to assess children's ability to interpret emotions through minimalist visual cues. A group of 100 children participated, each instructed to identify the emotion conveyed by each emoji. Researchers recorded their responses meticulously to gauge accuracy. Impressively, 71 children succeeded in identifying all six emotions accurately, achieving a perfect score of 5 out of 5. However, 29 children correctly recognized four out of the six emotions, showing some difficulty with particular expressions. This performance resulted in an overall average score of 4.71 out of 5, highlighting a strong collective ability among the children to decode emotional expressions, with a minor variation in precision across the group. This high score reflects the children's overall skill in interpreting monochrome emotional symbols, despite subtle challenges faced by a portion of the participants.

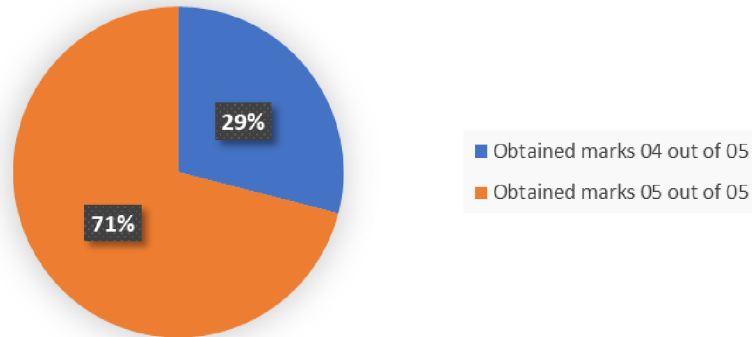


Figure-6. Coloring Emoji Identification Task Scores

5.7. Sally-Anne False Belief Task

In this study, researchers introduced the Sally-Anne False Belief Task by narrating the story and demonstrating the scenario to ensure participants grasped the concept. After this introduction, the children answered a series of five questions aimed at evaluating their theory of mind-specifically, their ability to recognize the beliefs and perspectives of others. Notably, 82 participants answered all five questions correctly, achieving a perfect score of 5 out of 5 and indicating a strong understanding of the task. Another 11 participants responded accurately to four questions, earning 4 out of 5, while 7 participants scored 3 out of 5. The overall performance score across participants averaged 4.75 out of 5 (or 95%), reflecting a high level of comprehension with only slight variability in responses across the group.

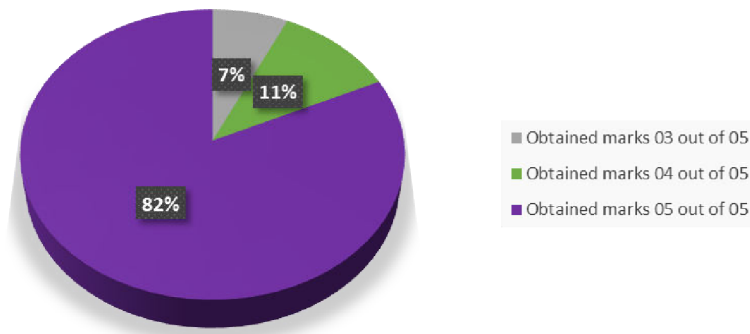


Figure-7: Sally-Anne False Belief Task Scores

5.8. Emotional Game

In this study, researchers presented six fragmented images, each portraying an individual displaying a distinct emotional expression, challenging participants to identify emotions from incomplete visual cues. A total of 100 children participated, with each child's responses carefully recorded to assess accuracy in emotion recognition. Remarkably, 91 children successfully identified all six emotions, achieving a perfect score of 5 out of 5, indicating a high level of interpretive skill. Meanwhile, 6 children correctly identified four emotions, resulting in a score of 4 out of 5, and an additional 3 children accurately identified three emotions, scoring 3 out of 5.

The combined performance yielded an overall average score of 4.88 out of 5, demonstrating that most children could effectively interpret emotional expressions even in fragmented images, with minimal variability in accuracy across the group.

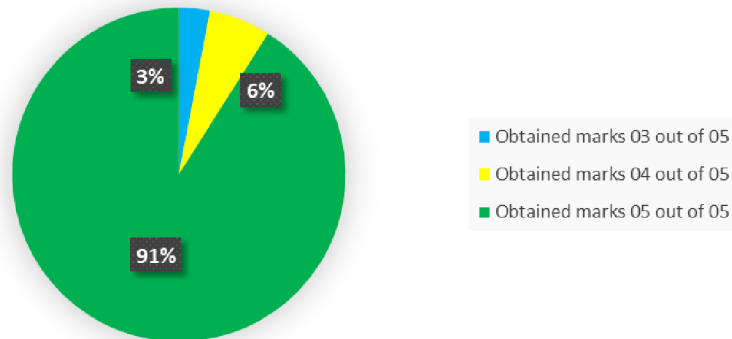


Figure-8: Emotional Game Task Scores

5.9. Emotional Sorting

In this study, 100 children were presented with 12 images, each depicting one of six distinct emotions, with two images dedicated to each emotion to test consistency in recognition. Researchers asked the children to identify the emotion in each picture and meticulously recorded their responses. Among the participants, 82 children demonstrated exceptional accuracy by correctly identifying all 12 images, achieving a perfect score of 5 out of 5. Another 12 children identified 10 images accurately, resulting in a score of 4 out of 5, while 6 children correctly recognized 8 images, earning a score of 3 out of 5. This distribution of results produced an overall average score of 4.76 out of 5, reflecting a strong general ability among the children to recognize emotions across multiple representations, with only minor variations in performance.

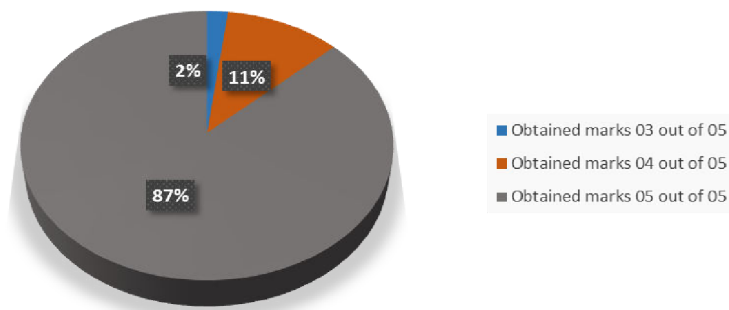


Figure-9: Emotional Sorting Task Scores

5.10. Comparative Insights

The assessment reveals advanced emotional recognition and interpretive abilities among Bangladeshi children aged 3–7, with consistently high scores across various tasks. The Personal Narrative Task achieved the highest average score of 4.92/5, showcasing exceptional emotional interpretation through storytelling. Similarly, the Pretend Play Task attained a flawless average of 5/5, highlighting superior imitation and engagement skills.

Visual tasks also demonstrated notable proficiency. The Picture Identification Task recorded an average score of 4.85/5, indicating reliable recognition of basic emotions. The Emoji Identification Task and Coloring Emoji Identification Task achieved averages of 4.78/5 and 4.71/5, respectively, with minor challenges in identifying abstract symbols. The Sally-Anne False Belief Task, assessing theory of mind, scored 4.75/5, reflecting strong perspective-taking capabilities. The Emotional Game Task reached an impressive average of 4.88/5, confirming accurate emotional interpretation even with fragmented visual cues. The Emotional Sorting Task produced an average score of 4.76/5, demonstrating consistent recognition across diverse representations.

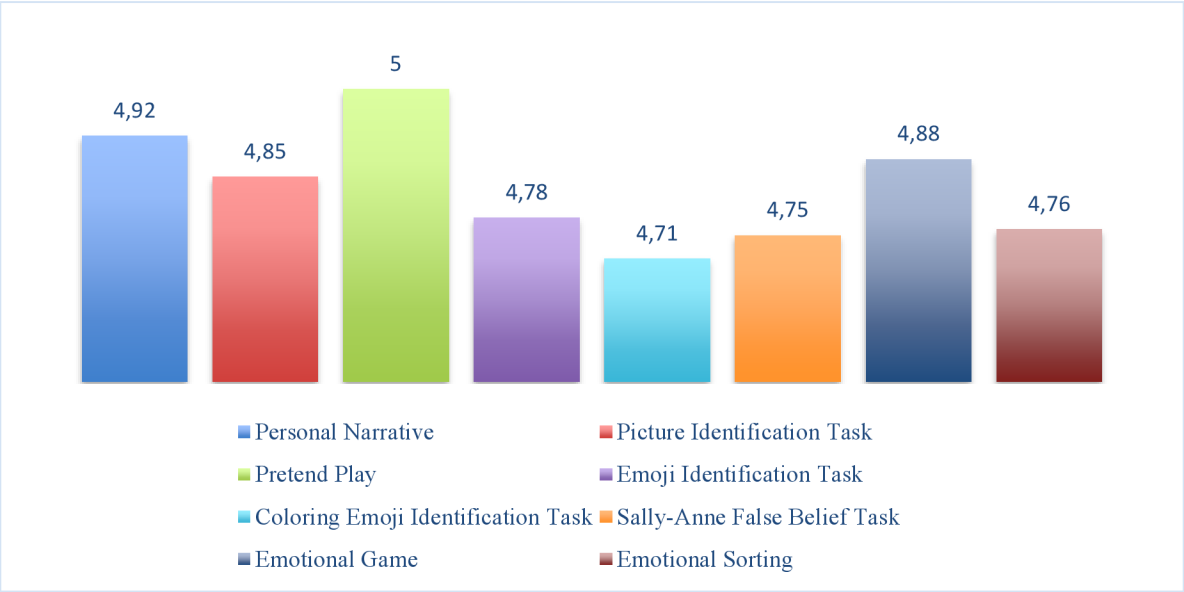


Figure-10: Overall Findings and Scores

The overall findings reveal exceptionally well-developed emotional recognition, interpretive abilities, and theory of mind among typically developing Bangladeshi children aged 3-7. The impressive cumulative average score of 38.65 out of 40 across all tasks underscores their advanced capacity to accurately identify, comprehend, and interpret emotional cues presented through various modalities, including narratives, visuals, and abstract representations.

The children's performance consistently demonstrated a sophisticated ability to recognize emotions and understand perspectives, indicating well-developed emotional and cognitive processing skills. Notably, minimal difficulties were observed, primarily when tasks involved abstract or minimalist cues, suggesting that their interpretive skills are more robust when dealing with concrete or structured stimuli. These findings collectively affirm the children's advanced emotional and cognitive competencies, highlighting their remarkable aptitude for comprehending and responding to complex emotional and social information.

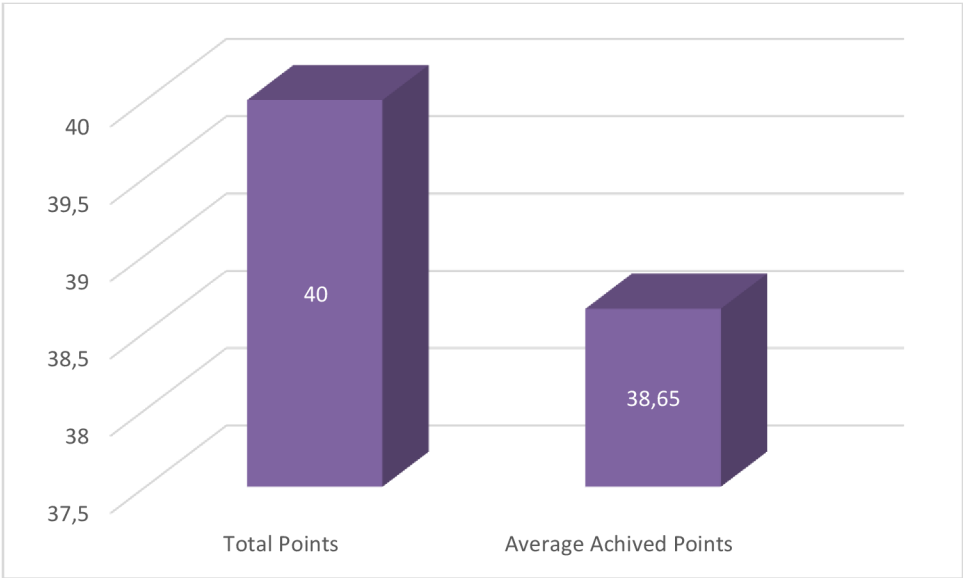


Figure-10: Average Achieved Points and Total Points

5.11. Discussion Summary

The findings of this study provide significant insights into the development of emotional recognition skills among typically developing Bangladeshi children, contributing novel perspectives to the existing body of knowledge with particular cultural and contextual relevance. Previous research underscores the critical role of emotional recognition in early childhood, linking it to subsequent social, psychological, and academic success (Bedford et al., 2021; Negrão et al., 2021). The present study's results are consistent with these findings, demonstrating that Bangladeshi children generally exhibit high accuracy in identifying emotions across various tasks, though some variability was observed across individual tasks and age groups. Notably, the outcomes align with Denham's (2009) proposition that early emotional education fosters emotional competency, as evidenced by the participants' strong performance in tasks like Personal Narrative, Picture Identification, and Pretend Play. This proficiency aligns with Bedford et al. (2021), who emphasize the significance of structured social interactions, suggesting that even within the collectivist framework of Bangladesh, children benefit considerably from culturally shaped social interactions and familial dynamics.

Furthermore, the study underscores the influence of cultural factors such as communal values and familial support in enhancing children's interpretive abilities, supporting the relevance of Bowlby's Attachment Theory (Bowlby, Ainsworth, & Bretherton, 1992), which links emotional recognition to the security of early attachments. Novel findings emerged from this study, indicating that while most children exhibited consistent competency in tasks involving complete images, a subset of participants demonstrated mild difficulty with tasks demanding more nuanced interpretation, particularly the Emoji Identification Task and Sally-Anne False Belief Task. This suggests that while children aged 3 to 7 years generally possess a foundational understanding of emotions, they may require additional support in deciphering subtler social cues, as suggested by Golan et al. (2006) in the context of interventions for children with Autism Spectrum Disorder.

Moreover, tasks requiring fragmented or abstract emotional cues, such as the Emotional Game, yielded slightly lower performance scores, implying the developmental challenges associated with processing incomplete information. This observation may be attributed to cognitive developmental differences within this age range, corresponding to Piaget's (1952) stages of cognitive development. Additionally, socioeconomic factors appeared to play a role, as children from diverse backgrounds displayed varying scores, supporting Vygotsky's (1978) assertion that the social environment significantly shapes cognitive growth. Considering the varied socioeconomic conditions within Bangladesh, which impact children's access to educational and emotional resources, the findings highlight the necessity for culturally adapted, inclusive educational strategies.

The limitations of this study must be acknowledged, including a limited timeframe, a relatively small sample size of 100 children, and the lack of advanced data processing tools such as Python, SPSS, and R, which may have restricted the depth of data analysis. Future research should aim to overcome these limitations by employing larger, more diverse samples and utilizing sophisticated statistical techniques to enhance the reliability and generalizability of the findings. Based on these insights, it is recommended that emotional education interventions be tailored to reflect cultural values and socioeconomic diversity, ensuring equitable access to emotional development support for all children. Additionally, introducing structured emotional literacy programs within early childhood education settings could address existing gaps, enhancing children's capacity to navigate complex emotional landscapes effectively. Overall, this study not only reinforces findings from predominantly Western-centric research but also offers a culturally nuanced perspective on emotional recognition, emphasizing both universal and culturally specific aspects of emotional development.

6. Conclusion

This research on the ability of typically developing Bangladeshi children, aged 3 to 7, to decipher emotional expressions provides key insights into their emotional recognition and interpretive skills. The study yielded consistently high scores across a variety of tasks, demonstrating that these children possess advanced abilities in emotional understanding. The findings, detailed point by point, reveal the following:

- **Personal Narrative Task:** A remarkable 92 out of 100 children achieved a perfect score of 5 out of 5, resulting in an average score of 4.92. This suggests that the children excel in interpreting emotions conveyed through narrative prompts, reflecting a high level of emotional interpretive skill.
- **Picture Identification Task:** 85 children successfully identified all six basic emotions, achieving an average score of 4.85. This underscores their proficiency in recognizing emotional cues, with minimal variation across the group, indicating reliability in emotion recognition.

- **Pretend Play Task:** Every child replicated the emotional actions demonstrated, yielding a perfect score of 5 out of 5. This outcome highlights the children's excellent engagement and ability to imitate emotional expressions, which are crucial components of social-emotional development.
- **Emoji Identification Task:** While 78 children achieved a perfect score, with an average of 4.78, a few children exhibited slight difficulty with specific emojis. This suggests that while emotional recognition via emoji-based symbols is generally strong, certain abstract representations may present minor challenges.
- **Coloring Emoji Identification Task:** In this task, where children identified emotions using minimalist monochrome emojis, 71 children scored perfectly, with an average score of 4.71. Although the task presented more abstract symbols, the children demonstrated strong interpretive abilities, despite small challenges.
- **Sally-Anne False Belief Task:** With a high average score of 4.75, this task measured the children's theory of mind—specifically, their ability to understand others' perspectives. The high performance indicates that the majority of the children were able to attribute emotions and mental states to others, a critical aspect of emotional intelligence.
- **Emotional Game Task:** In this task, where emotions were identified from fragmented images, 91 children achieved a perfect score, resulting in a high average of 4.88. This suggests that the children excel at identifying emotions even when the visual cues are incomplete or unclear, showcasing their cognitive flexibility and adaptability.
- **Emotional Sorting Task:** The children's performance in this task, where they identified emotions across 12 different images, yielded an average score of 4.76, reflecting consistent emotional recognition across multiple representations. This further reinforces their ability to decode emotional expressions in diverse formats.

Collectively, these findings indicate that the children in this study demonstrated highly developed emotional recognition and interpretation abilities. Their performance, with an overall score of 38.65 out of 40, is indicative of their strong emotional understanding, which aligns with typical developmental milestones in emotional intelligence. Minor variations in performance were observed in tasks that involved more abstract or minimalist emotional cues, but these differences were limited in scope and do not undermine the overall high levels of emotional competence.

These results have significant implications for understanding emotional development in early childhood. The ability to accurately decode and interpret emotions is fundamental for successful social interactions and emotional regulation. The children's high performance across a variety of tasks underscores the critical role that both visual and narrative-based cues play in fostering emotional comprehension in early childhood. Furthermore, the study highlights the importance of considering the varying levels of abstraction in emotional cues, as minor challenges were observed in tasks that involved more minimalist or fragmented emotional symbols.

From a practical standpoint, these findings hold great potential for informing educational practices and interventions aimed at enhancing emotional intelligence in young children. Early educational programs that incorporate diverse tasks—such as narrative storytelling, visual recognition, and pretend play—can provide rich opportunities for developing emotional understanding. Additionally, the study's results suggest that interventions focused on refining children's ability to interpret abstract emotional cues could further enhance their emotional intelligence, especially in contexts where minimalist or symbolic representations are prevalent.

In conclusion, the study affirms that typically developing Bangladeshi children in the age range of 3 to 7 possess advanced emotional recognition and interpretive abilities, as evidenced by their high scores across multiple tasks. These findings provide a strong foundation for future research into the factors influencing emotional development in diverse cultural and educational contexts, as well as the potential applications of such knowledge in fostering emotional competencies in early childhood education.

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